

When Are Triangles Invisible? Fundamental Limits of Higher-Order Structure Identifiability from Edge Flows

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Abstract—Topological signal processing models higher-order interactions by placing signals on the triangles of a simplicial complex, but in many applications the set of “filled” triangles is unknown and must be inferred from edge-flow observations. Existing work provides algorithms for this inference; we instead ask when it is *possible at all*. We show that the latent higher-order structure is observed only through the *curl* of the edge flows, which annihilates the gradient and harmonic parts of the signal. This reduces per-triangle detection to a two-variance Gaussian test whose optimal error exponent is a Gaussian Chernoff information, and yields a sharp *curl-invisibility* phase: when the curl signal-to-noise ratio ρ falls below $\rho^*(T) = \Theta(\sqrt{\log(1/\delta)/T})$, no estimator can recover the structure from T snapshots. A geometry-aware whitening of the curl statistic decorrelates confusable triangles and attains an effective per-triangle SNR $\rho_\tau^{\text{eff}} = \sigma_c^2/(\sigma_n^2(G^+)_{\tau\tau})$ governed by the triangle Gram matrix; we prove and empirically confirm a matching finite-sample recovery law. Finally we expose a second, SNR-independent obstruction: on a complete graph K_n the curl subspace has dimension $\binom{n-1}{2}$ while there are $\binom{n}{3}$ candidate triangles, so a $3/n$ fraction is identifiable. On real daily foreign-exchange flows both obstructions bind: the arbitrage-free market is curl-free to machine precision, and its complete currency graph exposes only a third of its triangles. All theorems are validated against Monte-Carlo simulation; code and data are released for full reproducibility.

Index Terms—Topological signal processing, simplicial complexes, Hodge decomposition, identifiability, detection theory, edge flows.

I. INTRODUCTION

Graph signal processing captures pairwise structure, but many networked measurements—currency exchange, traffic circulation, neural flow, statistical ranking—carry *higher-order* structure that lives on the triangles of a simplicial complex [1], [2], [3]. Topological signal processing (TSP) has rapidly built a toolbox on the Hodge Laplacian: convolutional filters [4], adaptive estimation [5], unrolled reconstruction [6], and inference of the complex itself from data [7], [8]. A recurring primitive is that the set of filled triangles is unknown and must be learned from edge-flow signals.

The existing inference methods are *algorithmic*: greedy or sparse-recovery procedures that return an estimate. None characterizes when the task is information-theoretically solv-

able. This paper supplies the missing *fundamental limits*. Our contributions are:

- 1) A *curl-annihilation* reduction (Lemma 1) showing that the latent structure is observable only through the flow’s curl, turning detection into a two-variance Gaussian test whose error exponent is a Gaussian Chernoff information (Prop. 1).
- 2) A sharp *curl-invisibility* phase transition (Theorem 1): the structure is unrecoverable below $\rho^*(T) = \Theta(\sqrt{\log(1/\delta)/T})$, with $C(\rho) \sim \rho^2/16$.
- 3) A *geometry-aware* whitened detector (Theorem 2) with a matching finite-sample recovery law governed by an effective SNR ρ_τ^{eff} ; it strictly dominates naive curl-energy thresholding under edge-sharing confusability.
- 4) An SNR-independent *rank obstruction* (Theorem 3): on K_n only a $3/n$ fraction of candidate triangles is identifiable at any SNR.

Every result is validated against Monte-Carlo simulation, and both obstructions are illustrated on real foreign-exchange flows.

II. BACKGROUND AND SIGNAL MODEL

A. Hodge structure

Fix a graph $(\mathcal{V}, \mathcal{E})$ oriented so that the node–edge incidence is $B_1 \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$ and, for a chosen set of filled triangles, the edge–triangle incidence is $B_2 \in \mathbb{R}^{|\mathcal{E}| \times |\mathcal{T}|}$. The boundary-of-boundary identity $B_1 B_2 = \mathbf{0}$ holds, and the Hodge 1-Laplacian $L_1 = B_1^\top B_1 + B_2 B_2^\top$ induces the orthogonal decomposition

$$\mathbb{R}^{|\mathcal{E}|} = \underbrace{\text{im } B_1^\top}_{\text{gradient}} \oplus \underbrace{\text{im } B_2}_{\text{curl}} \oplus \underbrace{\ker L_1}_{\text{harmonic}}. \quad (1)$$

The *curl* of an edge flow is $\text{curl } \mathbf{f} = B_2^\top \mathbf{f}$; a flow induced by a triangle signal $\mathbf{y}$ is $B_2 \mathbf{y}$.

B. Generative model

The 1-skeleton is known; the latent object is the set $\mathcal{S}$ of filled triangles among a candidate set $\mathcal{T}$ (e.g. all 3-cliques). We observe T i.i.d. edge-flow snapshots

$$\mathbf{f}_t = B_1^\top \mathbf{a}_t + B_{2,\mathcal{S}} \mathbf{y}_t + \mathbf{h}_t + \mathbf{n}_t, \quad t = 1, \dots, T, \quad (2)$$

with node potentials $\mathbf{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I})$ (gradient nuisance), triangle signals $\mathbf{y}_t \sim \mathcal{N}(\mathbf{0}, \sigma_c^2 \mathbf{I})$, harmonic nuisance $\mathbf{h}_t$ and noise $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$. $\mathbf{B}_{2,S}$ keeps the columns of the active triangles. The goal is to recover $\mathcal{S}$ from $\{\mathbf{f}_t\}$.

III. FUNDAMENTAL LIMITS

A. Curl annihilation and the single-triangle test

Lemma 1 (Curl annihilation). *For any $\mathbf{f}$, $\mathbf{B}_2^\top \mathbf{f}$ depends only on the curl component: $\mathbf{B}_2^\top (\mathbf{B}_1^\top \mathbf{a}) = \mathbf{0}$ and $\mathbf{B}_2^\top \mathbf{h} = \mathbf{0}$ for $\mathbf{h} \in \ker \mathbf{L}_1$. Hence under (2), $\mathbf{B}_2^\top \mathbf{f}_t = (\mathbf{B}_2^\top \mathbf{B}_{2,S}) \mathbf{y}_t + \mathbf{B}_2^\top \mathbf{n}_t$.*

Proof. $\mathbf{B}_1 \mathbf{B}_2 = \mathbf{0}$ gives $\mathbf{B}_2^\top \mathbf{B}_1^\top = \mathbf{0}$. If $\mathbf{L}_1 \mathbf{h} = \mathbf{0}$ then $\mathbf{h}^\top \mathbf{B}_2 \mathbf{B}_2^\top \mathbf{h} \leq \mathbf{h}^\top \mathbf{L}_1 \mathbf{h} = 0$, so $\mathbf{B}_2^\top \mathbf{h} = \mathbf{0}$. $\square$

Thus the gradient and harmonic nuisances—however large—are removed by the curl map, and only the active triangles and (projected) noise remain.

For an isolated candidate triangle τ (edge-disjoint from other candidates) the scalar $c_{\tau,t} = \mathbf{B}_{2,\tau}^\top \mathbf{f}_t$ is zero-mean Gaussian with variance $v_0 = 3\sigma_n^2$ if $\tau \notin \mathcal{S}$ and $v_1 = 9\sigma_c^2 + 3\sigma_n^2$ if $\tau \in \mathcal{S}$, since each triangle column has squared norm 3.

Proposition 1 (Two-variance detection). *Detecting an isolated τ from T snapshots is a two-variance Gaussian test. The energy $E = \sum_t c_{\tau,t}^2$ is a sufficient statistic with $E/v_i \sim \chi_T^2$ under H_i , and the minimum Bayes error decays as $\exp(-TC(v_0, v_1))$, where $C(v_0, v_1) = \max_{s \in [0,1]} \frac{1}{2} [\log(sw_0 + (1-s)w_1) - s \log w_0 - (1-s) \log w_1]$ is the Gaussian Chernoff information, $w_i = 1/v_i$.*

Theorem 1 (Curl-invisibility phase). *Let $\rho = 3\sigma_c^2/\sigma_n^2$, so $v_1 = v_0(1 + \rho)$. Then $C(\rho) \rightarrow 0$ as $\rho \rightarrow 0$ with $C(\rho) = \rho^2/16 + o(\rho^2)$. Consequently, for target error δ and budget T , reliable detection requires $\rho \geq \rho^*(T) = \Theta(\sqrt{\log(1/\delta)/T})$; below ρ^* the triangle is unrecoverable by any estimator.*

Proof sketch. C is the optimal Bayes error exponent (Prop. 1), so no test beats $\exp(-TC)$; a Taylor expansion of $C(\rho)$ at $\rho = 0$ gives the $\rho^2/16$ rate, whence $\exp(-TC(\rho)) \leq \delta$ forces $\rho = \Omega(\sqrt{\log(1/\delta)/T})$. $\square$

B. Support recovery and geometry-aware whitening

When candidate triangles are edge-disjoint, their curl scalars are independent, and a common Bayes threshold yields exact recovery with probability $\prod_{\tau \in \mathcal{S}} (1 - P_{\text{miss}}) \prod_{\tau \notin \mathcal{S}} (1 - P_{\text{fa}})$, a finite-sample law whose 50% contour is the phase boundary (Fig. 1).

Edge-sharing triangles are *confusable*: with the triangle Gram matrix $\mathbf{G} = \mathbf{B}_2^\top \mathbf{B}_2$ ($G_{\sigma\tau} = \pm 1$ when they share an edge), an active triangle leaks curl energy onto its inactive neighbours, so naive thresholding fails—and worsens as ρ grows. The remedy is to whiten.

Theorem 2 (Geometry-aware whitening). *Let $\hat{\mathbf{y}}_t = \mathbf{G}^+ \mathbf{B}_2^\top \mathbf{f}_t$. On the identifiable subspace, $\hat{\mathbf{y}}_{\tau,t} = y_{\tau,t} \mathbf{1}\{\tau \in \mathcal{S}\} + e_{\tau,t}$ with $\text{Cov}(e_t) = \sigma_n^2 \mathbf{G}^+$. Hence each triangle obeys a two-variance test with $v_{0,\tau} = \sigma_n^2 (\mathbf{G}^+)_{\tau\tau}$, $v_{1,\tau} = \sigma_c^2 + v_{0,\tau}$, and effective SNR $\rho_\tau^{\text{eff}} = \sigma_c^2 / (\sigma_n^2 (\mathbf{G}^+)_{\tau\tau})$. The resulting recovery probability factorizes over triangles.*

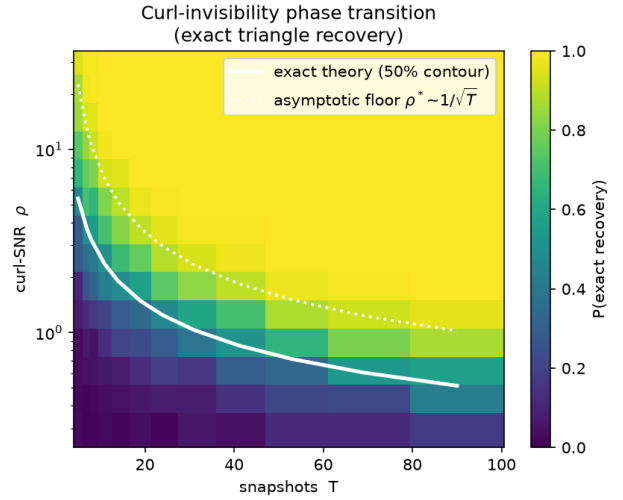


Fig. 1. Curl-invisibility phase transition for exact triangle recovery. Colour is empirical $P(\text{exact recovery})$; the solid line is the finite-sample theory, the dotted line the $1/\sqrt{T}$ asymptotic floor.

Proof sketch. By Lemma 1, $\mathbf{B}_2^\top \mathbf{f} = \mathbf{G}_{:,S} \mathbf{y} + \mathbf{B}_2^\top \mathbf{n}$. Applying $\mathbf{G}^+$ and using $\mathbf{G}^+ \mathbf{G}_{:,S} = \mathbf{I}_{:,S}$ on the identifiable subspace gives the mean; the noise covariance is $\sigma_n^2 \mathbf{G}^+ \mathbf{G} \mathbf{G}^+ = \sigma_n^2 \mathbf{G}^+$. $\square$

The factor $(\mathbf{G}^+)_{\tau\tau}$ is the geometry/confusability cost: $1/3$ for an isolated triangle, larger as curl signatures overlap.

C. The rank obstruction

Whitening cannot help when the curl signatures are linearly dependent.

Theorem 3 (Geometry-side invisibility). *$\mathcal{S}$ is identifiable only modulo ker-equivalence of $\mathbf{B}_2$: if $\text{im } \mathbf{B}_{2,S} = \text{im } \mathbf{B}_{2,S'}$ then $\mathcal{S}, \mathcal{S}'$ are indistinguishable at any SNR. On the complete graph K_n (all triangles filled), $\dim \text{im } \mathbf{B}_2 = \binom{n-1}{2}$ while the candidate count is $\binom{n}{3}$, so the identifiable fraction is exactly $3/n$.*

Proof. Only $\text{im } \mathbf{B}_2$ affects the flow, giving indistinguishability. For K_n the 2-skeleton is simply connected, so $\text{im } \mathbf{B}_2 = \ker \mathbf{B}_1$ (the cycle space) of dimension $|\mathcal{E}| - (n-1) = \binom{n-1}{2}$; dividing by $\binom{n}{3}$ gives $3/n$. $\square$

IV. EXPERIMENTS

All experiments run on CPU; every theoretical curve below is overlaid on independent Monte-Carlo simulation. Code and data reproduce all figures via a single script.

A. Phase transition

Fig. 1 sweeps (T, ρ) on an edge-disjoint planted complex with gradient and harmonic nuisance present. The empirical exact-recovery probability undergoes a sharp transition; the solid curve is the finite-sample theory of Theorem 2, whose empirical 50% contour it tracks to a median ratio of 1.01. The dotted curve is the asymptotic floor $\rho^* \sim 1/\sqrt{T}$ of Theorem 1.

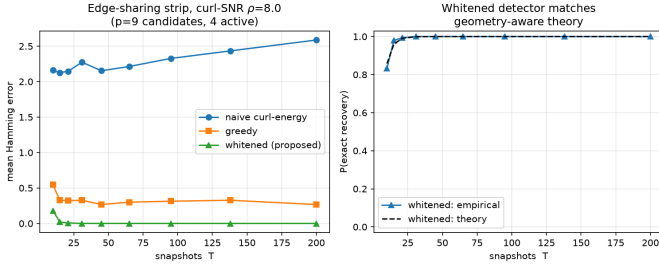


Fig. 2. Edge-sharing strip. Left: mean Hamming error vs. T ; the naive detector never recovers. Right: whitened detector matches the geometry-aware recovery law.

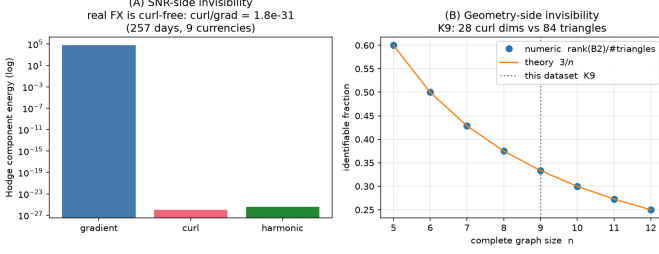


Fig. 3. Real FX flows. (A) SNR-side invisibility: curl energy is machine-zero relative to gradient. (B) Geometry-side invisibility: identifiable fraction $= 3/n$.

B. Confusability

On an edge-sharing triangle strip (Fig. 2), the naive curl-energy detector fails and *worsens* with ρ due to leakage, while the whitened detector of Theorem 2 recovers the support and matches the geometry-aware theory; a greedy baseline lies in between.

C. Real foreign-exchange flows

We form edge flows from 257 trading days of ECB daily reference rates over nine currencies (Fig. 3). Both obstructions bind: (A) an arbitrage-free market is a pure gradient, so its curl energy is 10^{-31} of its gradient energy—the higher-order structure is invisible at any budget; (B) the complete currency graph K_9 has only 28 curl dimensions for 84 candidate triangles, matching the $3/n$ law of Theorem 3.

V. CONCLUSION

We gave the first fundamental limits for identifying higher-order structure from edge flows: a curl-invisibility phase transition, a geometry-aware whitened detector that attains it, and a rank obstruction that no SNR can overcome. The results delimit when the algorithmic TSP inference methods can succeed, and are borne out on real data. Extending the converse to sub-Gaussian signals and to partial edge sampling are natural next steps.

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